



INFORMATION SECURITY GOVERNANCE AND RISK MANAGEMENT

MECR1023

SECTION 52

LECTURER

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SELF REFLECTION 01 (CHAPTER 1)

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A) Work Breakdown Structure (WBS)

Exercise: Developing a Work Breakdown Structure (WBS) for a Security Awareness Program

Scenario: As the project manager, you are responsible for implementing a comprehensive security awareness program at your organization. The program's goal is to educate employees on cybersecurity best practices, company policies, and effective responses to security threats.

a. Project Phases (Level 1)

- Planning phase – define the program's objectives, scope, and deliverables.
- Design and development phase – create and prepare the awareness materials and training resources.
- Implementation phase – deliver the training sessions and distribute learning materials.
- Evaluation phase – review and measure the program's success and employee comprehension.
- Closure phase – keep the program current through updates and ongoing awareness efforts.

b. Phase breakdown into tasks (Level 2)

1. Planning phase
 - 1.1 Determine clear goals and measurable outcomes.
 - 1.2 Identify stakeholders, team members, and assign responsibilities.
 - 1.3 Develop the overall schedule, resource plan, and communication structure.
2. Design and development phase
 - 2.1 Produce cybersecurity learning modules.
 - 2.2 Design engaging visual materials.
 - 2.3 Present the developed content for review and approval by management.
3. Implementation phase
 - 3.1 Plan and conduct employee awareness sessions.
 - 3.2 Monitor attendance and participation levels.
 - 3.3 Distribute awareness materials through internal communication channels and platforms.
4. Evaluation phase
 - 4.1 Administer post-training assessments and feedback surveys.
 - 4.2 Analyze results and identify areas for improvement.
 - 4.3 Document the overall findings and recommend enhancements.

- 5. Closure phase
 - 5.1 Revise and refresh content to reflect emerging security trends and policy changes.
 - 5.2 Schedule follow-up or refresher sessions periodically.
 - 5.3 Share updated progress and awareness outcomes with top management.

c. Task decomposition into subtasks (Level 3)

1.Planning phase

- 1.1 Determine clear goals and measurable outcomes.
 - 1.1.1 Review existing security incident reports to identify knowledge gaps.
 - 1.1.2 Define specific, measurable Key Performance Indicators (KPIs) like a target reduction in phishing click-through rates.
- 1.2 Identify stakeholders, team members, and assign responsibilities.
 - 1.2.1 Compile a list of key stakeholders from IT, HR, Legal, and senior management.
 - 1.2.2 Schedule a kick-off meeting to establish communication protocols.
- 1.3 Develop the overall schedule, resource plan, and communication structure.
 - 1.3.1 Draft a budget covering costs for materials, software, and potential external vendors.

2.Design and development phase

- 2.1 Produce cybersecurity learning modules
 - 2.1.1 Script the content for the phishing awareness module.
 - 2.1.2 Develop interactive password creation exercises.
 - 2.1.3 Create case study scenarios for data protection protocols.
- 2.2 Design engaging visual materials
 - 2.2.1 Draft storyboards for short, animated video explainers.
 - 2.2.2 Create infographics on topics like social engineering tactics.
 - 2.2.3 Design a series of posters highlighting key security "Dos and Don'ts".
- 2.3 Present the developed content for review and approval by management.
 - 2.3.1 Schedule and lead a presentation walkthrough of the program content.

3. Implementation phase

- 3.1 Plan and conduct employee awareness sessions.
 - 3.1.1 Secure training rooms or set up virtual meeting links for all scheduled sessions.
 - 3.1.2 Deliver the training sessions, incorporating real-time Q&A.
- 3.2 Monitor attendance and participation levels.
 - 3.2.1 Track employee sign-ins for all live sessions.
 - 3.2.2 Follow up with managers on employees who have not completed required training.
- 3.3 Distribute awareness materials through internal communication channels and platforms.

3.3.1 Send a company-wide email with links to key resources and videos.

4. Evaluation phase

4.1 Administer post-training assessments and feedback surveys.

4.1.1 Deploy a final quiz to test knowledge retention.

4.1.2 Distribute an anonymous survey to gather feedback on training quality and relevance.

4.2 Analyze results and identify areas for improvement.

4.2.1 Compile a list of specific content and delivery improvements for the next cycle.

4.3 Document the overall findings and recommend enhancements.

4.3.1 Create a comprehensive report detailing the program's performance against the initial KPIs.

4.3.2 Update the project archive with all evaluation data and the final report.

5. Closure phase

5.1 Revise and refresh content to reflect emerging security trends and policy changes.

5.2 Schedule follow-up or refresher sessions periodically.

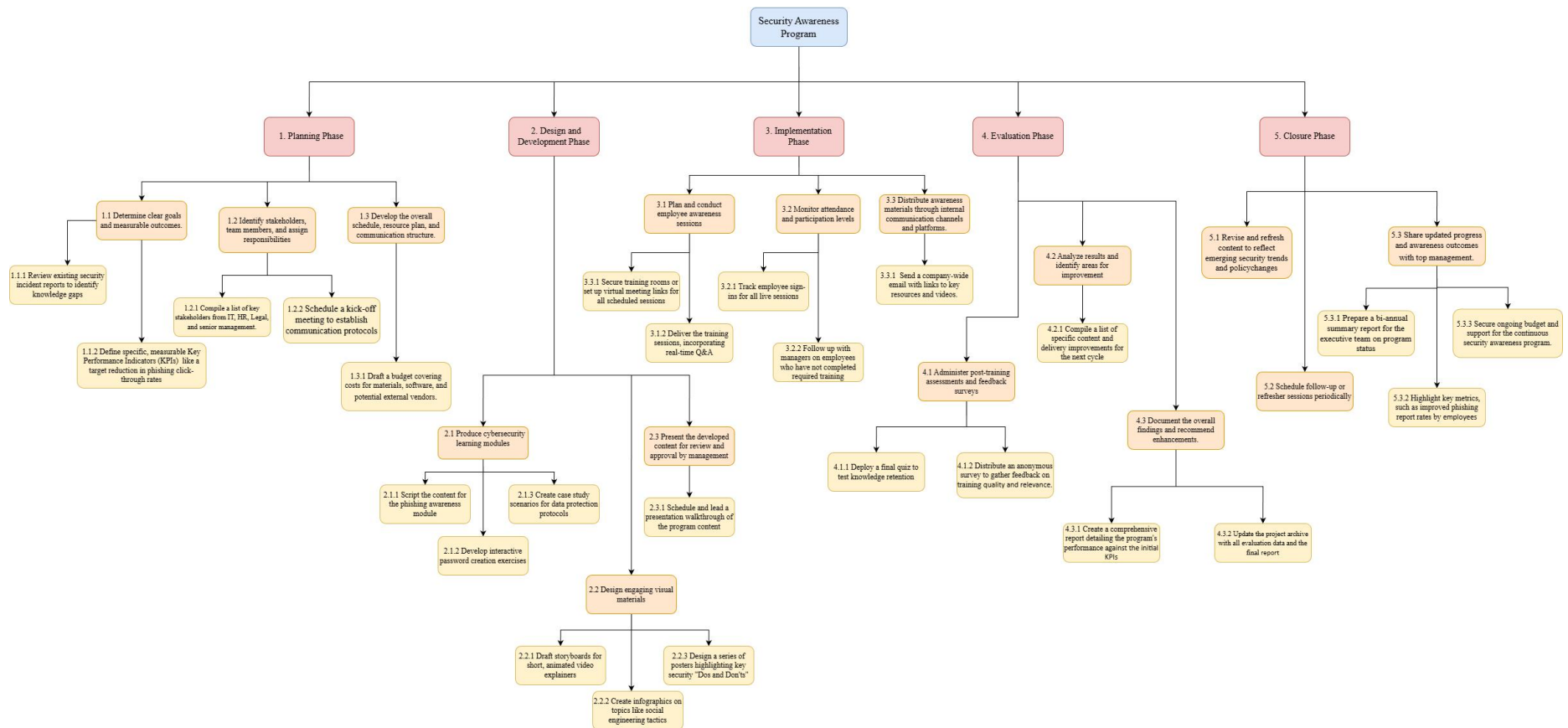
5.3 Share updated progress and awareness outcomes with top management.

5.3.1 Prepare a bi-annual summary report for the executive team on program status.

5.3.2 Highlight key metrics, such as improved phishing report rates by employees.

5.3.3 Secure ongoing budget and support for the continuous security awareness program.

d. Draw the WBS Diagram



e. Reflection

The process of constructing the Work Breakdown Structure presented distinct challenges, primarily in defining the project's boundaries and determining the optimal level of detail. A key difficulty was distinguishing between the finite implementation project and ongoing operational maintenance, ensuring the WBS remained focused on specific deliverables with a clear endpoint. Furthermore, striking a balance was necessary to make the WBS comprehensive enough for effective management without becoming bogusly granular, which would hinder its practical use. This decomposition also naturally revealed complex task interdependencies and prompted essential considerations regarding resource allocation and team skills for each work package.

Developing such a detailed WBS profoundly benefits the project's entire lifecycle. It transforms an abstract goal into a concrete plan, serving as the foundational framework for accurate scheduling, budgeting, and resource forecasting. This clarity prevents oversight of critical tasks and allows for proactive risk management. During implementation, the WBS establishes unambiguous accountability, enabling precise progress tracking against specific deliverables rather than vague milestones. It also acts as a vital communication tool, aligning all stakeholders on the project's scope to prevent scope creep and facilitates built-in quality control checks throughout each phase, ultimately increasing the likelihood of achieving the project's measurable objectives.

B) Task Sequencing Approach

a) Program Evaluation Review Technique (PERT)

Exercise: Task Sequencing for Security Policy Development

Scenario: You are tasked with creating a comprehensive security policy for your office, addressing key areas such as data protection, access control, incident response, and employee responsibilities. You will use the Program Evaluation and Review Technique (PERT) to sequence tasks, estimate timelines, and ensure the policy is completed on time.

Task	Description	O (days)	M (days)	P (days)
A: Define Security Objectives	Outline policy goals and scope	1	2	4
B: Conduct Risk Assessment	Analyze risks and vulnerabilities	2	3	5
C: Review Compliance	Review legal and regulatory needs	1	2	3
D: Draft Policy Sections	Write initial policy draft	3	4	6
E: Consult Stakeholders	Obtain feedback from stakeholders	2	3	5
F: Revise Draft	Make changes based on feedback	1	2	3
G: Approve Policy	Final approval from management	1	2	3
H: Implement Training	Prepare and deliver training	2	3	5
I: Monitor Compliance	Set up monitoring processes	1	2	3

1. Draw the simple network dependencies between tasks. For instance:

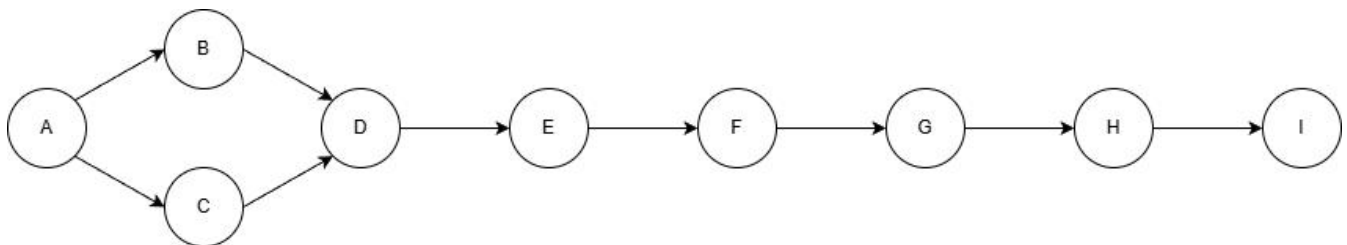
Task A must be completed before B, C, and D.

Task B and C must be completed before D.

Task D must be completed before E.

Task E must be completed before F.

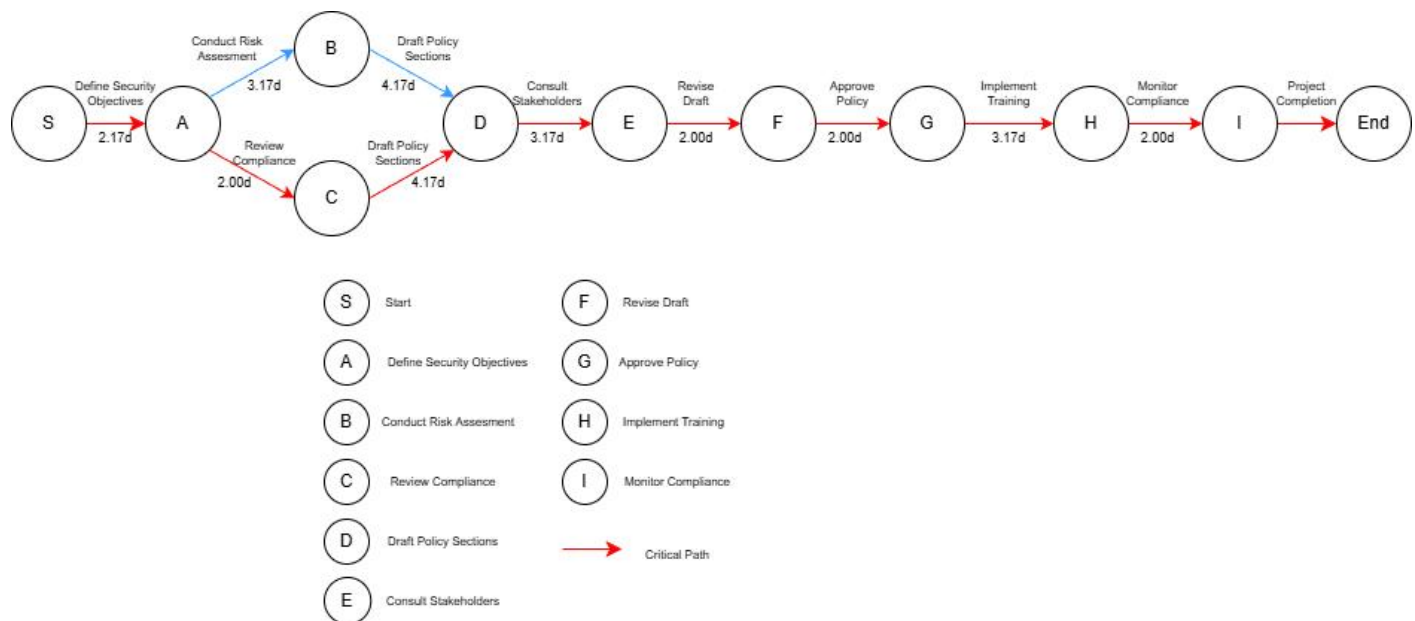
Task F must be completed before G, and so on.



2. Calculate Expected Time (TE) for Each Task from the table using PERT formula.

Task	O	M	P	TE = (O+4M+P)/6 (days)
A	1	2	4	$(1 + 4 \times 2 + 4)/6 = 2.17$
B	2	3	5	$(2 + 4 \times 3 + 5)/6 = 3.17$
C	1	2	3	$(1 + 4 \times 2 + 3)/6 = 2.00$
D	3	4	6	$(3 + 4 \times 4 + 6)/6 = 4.17$
E	2	3	5	$(2 + 4 \times 3 + 5)/6 = 3.17$
F	1	2	3	$(1 + 4 \times 2 + 3)/6 = 2.00$
G	1	2	3	$(1 + 4 \times 2 + 3)/6 = 2.00$
H	2	3	5	$(2 + 4 \times 3 + 5)/6 = 3.17$
I	1	2	3	$(1 + 4 \times 2 + 3)/6 = 2.00$

3. Create a PERT diagram showing the tasks (nodes), dependencies (arrows), and the calculated expected time for each task. Connect tasks based on their dependencies, ensuring that each task flows logically from start to finish.



4. Identify the Critical Path. Calculate the earliest start (ES), earliest finish (EF), latest start (LS), and latest finish (LF) for each task to identify the critical path. The critical path is the longest path through the network, showing tasks that, if delayed, will delay the entire project.

i) Path 1: $S \rightarrow A \rightarrow B \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H \rightarrow I \rightarrow \text{End}$
 $2.17 + 3.17 + 4.17 + 3.17 + 2.00 + 2.00 + 3.17 + 2.00 = \mathbf{21.85 \text{ days}}$

Path 2: $S \rightarrow A \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H \rightarrow I \rightarrow \text{End}$
 $2.17 + 2.00 + 4.17 + 3.17 + 2.00 + 2.00 + 3.17 + 2.00 = \mathbf{20.68 \text{ days}}$

Critical Path is Path 1 which is $A \rightarrow B \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H \rightarrow I$.

ii)

- Earliest Start (ES) and Earliest Finish (EF)

Rules:

ES (Earliest Start) = maximum EF of all predecessors

EF (Earliest Finish) = ES + Duration

Activity	Predecessor(s)	Duration	Formula	ES	EF = ES + Duration
A	—	2.17	Start $\rightarrow$ ES = 0	0.00	$0.00 + 2.17 = 2.17$
B	A	3.17	ES = EF(A)	2.17	$2.17 + 3.17 = 5.34$
C	A	2.00	ES = EF(A)	2.17	$2.17 + 2.00 = 4.17$
D	B, C	4.17	ES = $\max(\text{EF(B)}, \text{EF(C)}) = \max(5.34, 4.17)$	5.34	$5.34 + 4.17 = 9.51$
E	D	3.17	ES = EF(D)	9.51	$9.51 + 3.17 = 12.68$
F	E	2.00	ES = EF(E)	12.68	$12.68 + 2.00 = 14.68$
G	F	2.00	ES = EF(F)	14.68	$14.68 + 2.00 = 16.68$
H	G	3.17	ES = EF(G)	16.68	$16.68 + 3.17 = 19.85$
I	H	2.00	ES = EF(H)	19.85	$19.85 + 2.00 = 21.85$

Project duration = EF(I) = **21.85 days**

- Latest Start (LS) and Latest Finish (LF)

Rules:

LF (Latest Finish) = minimum LS of all successors

LS (Latest Start) = LF – Duration

Work backwards from project end = **21.85 days**

Activity	Successor(s)	Duration (TE days)	Formula	LF (days)	LS = LF – Duration (days)
I	—	2.00	Final → LF = Project End	21.85	21.85 – 2.00 = 19.85
H	I	3.17	LF = LS(I)	19.85	19.85 – 3.17 = 16.68
G	H	2.00	LF = LS(H)	16.68	16.68 – 2.00 = 14.68
F	G	2.00	LF = LS(G)	14.68	14.68 – 2.00 = 12.68
E	F	3.17	LF = LS(F)	12.68	12.68 – 3.17 = 9.51
D	E	4.17	LF = LS(E)	9.51	9.51 – 4.17 = 5.34
B	D	3.17	LF = min(LS of successors) = LS(D)	5.34	5.34 – 3.17 = 2.17
C	D	2.00	LF = LS(D)	5.34	5.34 – 2.00 = 3.34
A	B, C	2.17	LF = min(LS(B),LS(C)) = min(2.17,3.34)	2.17	2.17 – 2.17 = 0.00

- Slack (Float) and Critical Path

Rules: **Slack = LS – ES** (equivalently LF – EF). If slack = 0 → activity is critical.

Duration (TE)	ES	EF	LS	LF	Slack = LS – ES	Critical?
A	2.17	0.00	2.17	0.00	2.17	0.00
B	3.17	2.17	5.34	2.17	5.34	0.00
C	2.00	2.17	4.17	3.34	5.34	1.17
D	4.17	5.34	9.51	5.34	9.51	0.00
E	3.17	9.51	12.68	9.51	12.68	0.00
F	2.00	12.68	14.68	12.68	14.68	0.00
G	2.00	14.68	16.68	14.68	16.68	0.00
H	3.17	16.68	19.85	16.68	19.85	0.00
I	2.00	19.85	21.85	19.85	21.85	0.00

- Activities with zero slack form the critical path.
- Critical path: $A \rightarrow B \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H \rightarrow I$
- Total project time (sum along critical path) = 21.85 days.

C has slack = 1.17 days (it can be delayed up to 1.17 days without delaying the project). All other tasks have zero slack (critical).

5. Calculate Total project time = sum of TE for tasks on the critical path.

Critical Path: $S \rightarrow A \rightarrow B \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H \rightarrow I \rightarrow \text{End}$

Total Project Time = $2.17 + 3.17 + 4.17 + 3.17 + 2.00 + 2.00 + 3.17 + 2.00 = \mathbf{21.85 \text{ days}}$

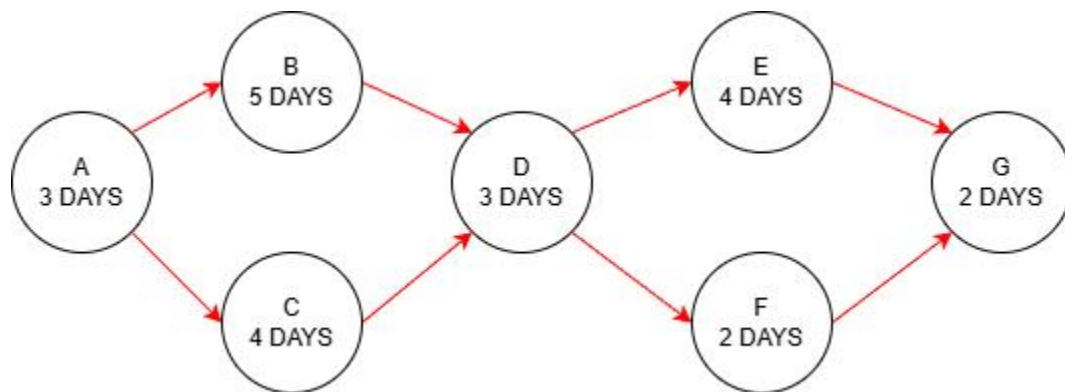
Total Project Duration = 21.68 days (≈ 22 days)

b) Critical Path Method (CPM)

As a manager in a security company, you are tasked with overseeing a project aimed at implementing an integrated security system for a corporate building. The project includes installing surveillance equipment, access control, alarm systems, and conducting security training for the staff. Each task has dependencies, durations, and specific requirements. Your role is to ensure that the project is completed on schedule by identifying the critical path and assessing the slack time for each activity.

Activity	Description	Duration (days)	Predecessor(s)
A	Site Assessment and Planning	3	-
B	Surveillance System Installation	5	A
C	Access Control System Setup	4	A
D	Alarm System Installation	3	B, C
E	System Integration and Testing	4	D
F	Staff Security Training	2	D
G	Final Inspection and Handover	2	E, F

1. Create a Project Network Diagram:
 - Construct a network diagram based on the project activities and dependencies.



2. Identify the Critical Path:
 - Calculate the total duration for each path in the network diagram and identify the critical path, which represents the longest sequence of dependent activities.

Path	Formula	Total Duration (days)
1. A → B → D → E → G	3 + 5 + 3 + 4 + 2	17 days
2. A → C → D → E → G	3 + 4 + 3 + 4 + 2	16 days
3. A → B → D → F → G	3 + 5 + 3 + 2 + 2	15 days
4. A → C → D → F → G	3 + 4 + 3 + 2 + 2	14 days

The longest total duration = 17 days

The critical path is A → B → D → E → G, with total duration of 17 days.

3. Calculate Early and Late Start/Finish Times:
 - For each activity, determine the:

i. Earliest Start (ES) and Earliest Finish (EF)

Forward pass → (ES/EF)

- Rule:

ES (start task) = 0

ES (activity) = max (EF of all predecessors)
EF = ES + Duration

- Calculation for Forward pass (ES/EF)

▪ A: ES = 0, EF = 0 + 3 = 3

▪ B: ES = EF(A) = 3, EF = 3 + 5 = 8

▪ C: ES = EF(A) = 3, EF = 3 + 4 = 7

- D: $ES = \max (EF B, EF C) = \max (8,7) = 8$, $EF = 8 + 3 = 11$
- E: $ES = EF(D) = 11$, $EF = 11 + 4 = 15$
- F: $ES = EF(D) = 11$, $EF = 11 + 2 = 13$
- G: $ES = \max (EF E, EF F) = \max (15,13) = 15$, $EF = 15 + 2 = 17$

ii. Latest Start (LS) and Latest Finish (LF)

Backward pass $\rightarrow$ LS & LF

- Rule:
 - LF (finish task) = EF (finish task)
 - (project duration) LF (activity) = min
 - (LS of all immediate successors) LS =
 - LF – Duration
- Calculation for Backward pass (LS/LF)
 - G: $LF = 17$, $LS = 17 - 2 = 15$
 - E: $LF = LS(G) = 15$, $LS = 15 - 4 = 11$
 - F: $LF = LS(G) = 15$, $LS = 15 - 2 = 13$
 - D: $LF = \min (LS E, LS F) = \min (11,13) = 11$, $LS = 11 - 3 = 8$
 - B: $LF = LS(D) = 8$, $LS = 8 - 5 = 3$
 - C: $LF = LS(D) = 8$, $LS = 8 - 4 = 4$
 - A: $LF = \min (LS B, LS C) = \min (3,4) = 3$, $LS = 3 - 3 = 0$

Activity	Duration	ES	EF	LS	LF
A	3	0	3	0	3
B	5	3	8	3	8
C	4	3	7	4	8
D	3	8	11	8	11
E	4	11	15	11	15
F	2	11	13	13	15
G	2	15	17	15	17

4. Determine Slack Time:

- Calculate the slack time (if any) for each activity to identify tasks that can be delayed without affecting the project completion date.

$$\text{Slack} = \text{LS} - \text{ES} \text{ or } \text{Slack} = \text{LF} - \text{EF}$$

Activity	Duration (Days)	ES	EF	LS	LF	Slack (LS-ES)
A	3	0	3	0	3	0
B	5	3	8	3	8	0
C	4	3	7	4	8	1
D	3	8	11	8	11	0
E	4	11	15	11	15	0
F	2	11	13	13	15	2
G	2	15	17	15	17	0

- Activities A, B, D, E, and G have zero slack, indicating they are on the critical path.
- Activities C (1 day) and F (2 days) have positive slack, which means they can be delayed by that number of days without impacting the total project duration.
- The critical path (A → B → D → E → G) determines the overall project completion time of 17 days.

5. Analysis and Reflection:

- Provide a brief analysis of why certain tasks are on the critical path and how managing slack time could impact the project timeline.

The tasks on the critical path (A → B → D → E → G) are those whose durations and dependencies directly determine the project's finish date because any delay in them immediately delays the whole project; they form the longest chain of dependent work with zero slack. In contrast, activities C and F have limited slack (1 and 2 days) and can be delayed slightly without affecting the completion date, which makes them useful buffers for handling minor issues or for leveling resources. Managing the critical path means prioritizing resources, monitoring progress more frequently, and protecting those activities from interruptions. If you need to shorten the schedule, focus on reducing durations or overlaps on critical activities (through crashing or fast-tracking) rather than non-critical ones. Conversely, if a delay is unavoidable in a non-critical task, you can often reassign its resources temporarily to critical tasks to prevent a project slip. Finally, keeping contingency time and clear communication channels around critical-path tasks reduces surprises and gives the project manager the best leverage to keep the entire implementation on schedule.

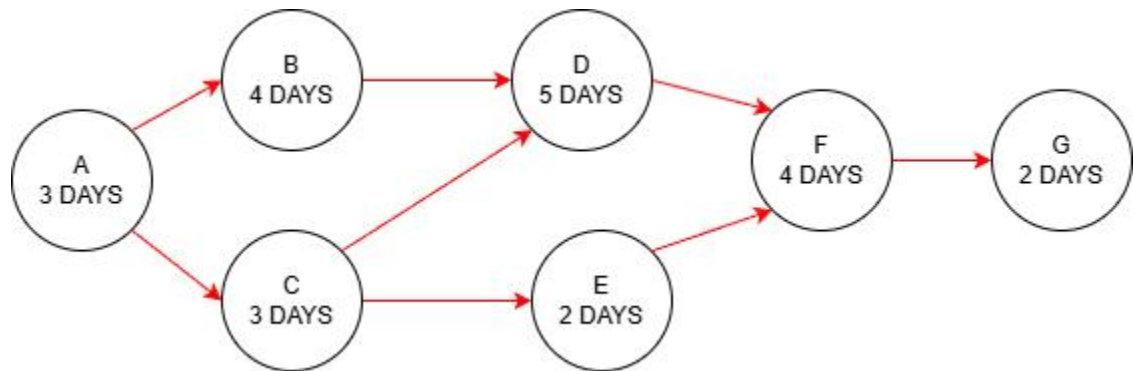
c) Critical Path Analysis

As a manager in a security company, you are tasked with supervising a network security project for a large organization. The project includes tasks such as installing firewalls, deploying intrusion detection systems (IDS), setting up virtual private networks (VPNs), and conducting penetration testing. Your objective is to ensure timely project completion by applying Critical Path Analysis to determine the critical path and evaluate slack time for each task.

Activity	Description	Duration (days)	Predecessor(s)
A	Requirements Analysis	3	-
B	Firewall Installation	4	A
C	VPN Configuration	3	A
D	IDS Setup	5	B, C
E	Security Policy Implementation	2	B
F	Penetration Testing	4	D, E
G	Final Review and Documentation	2	F

1. Draw a Project Network Diagram:

- Construct a network diagram illustrating each activity and its dependencies.



2. Identify the Critical Path:

- Calculate each possible path's duration in the network and identify the critical path, representing the longest sequence of dependent activities.

Path	Duration Formula	Total Duration (Days)
A → B → D → F → G	3 + 4 + 5 + 4 + 2	18
A → C → D → F → G	3 + 3 + 5 + 4 + 2	17
A → B → E → F → G	3 + 4 + 2 + 4 + 2	15

The longest total duration = 18 days

3. Calculate Early and Late Start/Finish Times:

- For each activity, calculate:

- Earliest Start (ES) and Earliest Finish (EF)

Rules:

ES (Earliest Start) = maximum EF of all predecessors

EF (Earliest Finish) = ES + Duration

Activity	Predecessor(s)	Duration	Formula	ES	EF = ES + Duration
A	—	3	Start activity → ES = 0	0	0 + 3 = 3
B	A	4	ES = EF(A)	3	3 + 4 = 7
C	A	3	ES = EF(A)	3	3 + 3 = 6
D	B, C	5	ES = max (EF(B), EF(C)) = max (7, 6)	7	7 + 5 = 12
E	B	2	ES = EF(B)	7	7 + 2 = 9
F	D, E	4	ES = max (EF(D), EF(E)) = max (12, 9)	12	12 + 4 = 16
G	F	2	ES = EF(F)	16	16 + 2 = 18

- Latest Start (LS) and Latest Finish (LF)

Rules:

LF (Latest Finish) = minimum LS of all successors

LS (Latest Start) = LF – Duration

Activity	Successor(s)	Duration	Formula	LF	LS = LF – Duration
G	—	2	Final activity → LF = Project End = 18	18	18 – 2 = 16
F	G	4	LF = LS(G) = 16	16	16 – 4 = 12
D	F	5	LF = LS(F) = 12	12	12 – 5 = 7
E	F	2	LF = LS(F) = 12	12	12 – 2 = 10
B	D, E	4	LF = min (LS(D), LS(E)) = min (7, 10)	7	7 – 4 = 3
C	D	3	LF = LS(D) = 7	7	7 – 3 = 4
A	B, C	3	LF = min (LS(B), LS(C)) = min (3, 4)	3	3 – 3 = 0

4. Determine Slack Time:

- Calculate slack time for each activity, identifying non-critical tasks that can be delayed without affecting the project's overall timeline.

$$\text{Slack} = \text{LS} - \text{ES (or)} \text{LF} - \text{EF}$$

Activity	Duration	ES	EF	LS	LF	Slack
A	3	0	3	0	3	0
B	4	3	7	3	7	0
C	3	3	6	4	7	1
D	5	7	12	7	12	0
E	2	7	9	10	12	3
F	4	12	16	12	16	0
G	2	16	18	16	18	0

The critical path for this project is $A \rightarrow B \rightarrow D \rightarrow F \rightarrow G$, taking 18 days to complete. These tasks are critical because they depend directly on each other, and any delay in one will delay the whole project. Activities C and E have slack time (1 and 3 days), meaning they can be delayed slightly without affecting the final deadline. Managing slack is important because it allows flexibility in scheduling and helps focus resources on the critical tasks to prevent delays. By monitoring the critical path closely and using slack wisely, the project can stay on track and be completed on time.